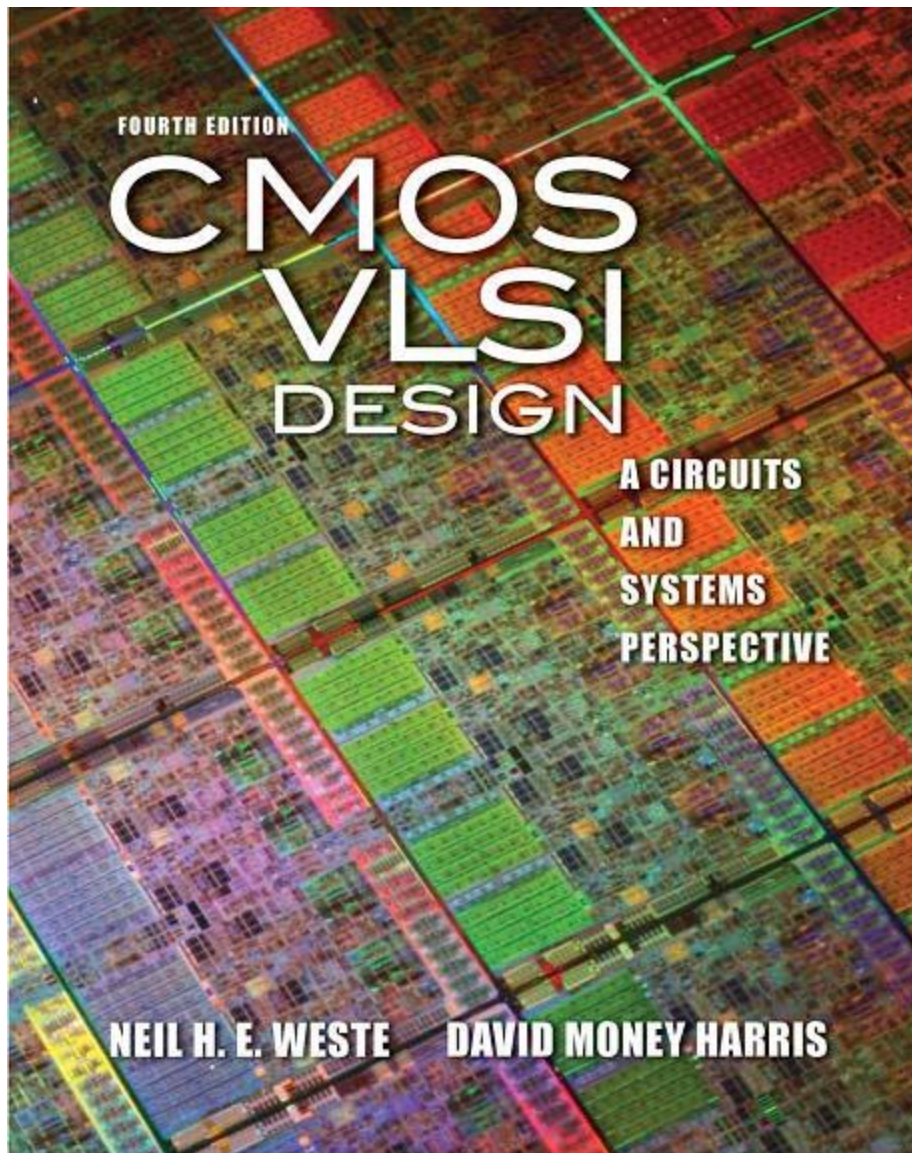
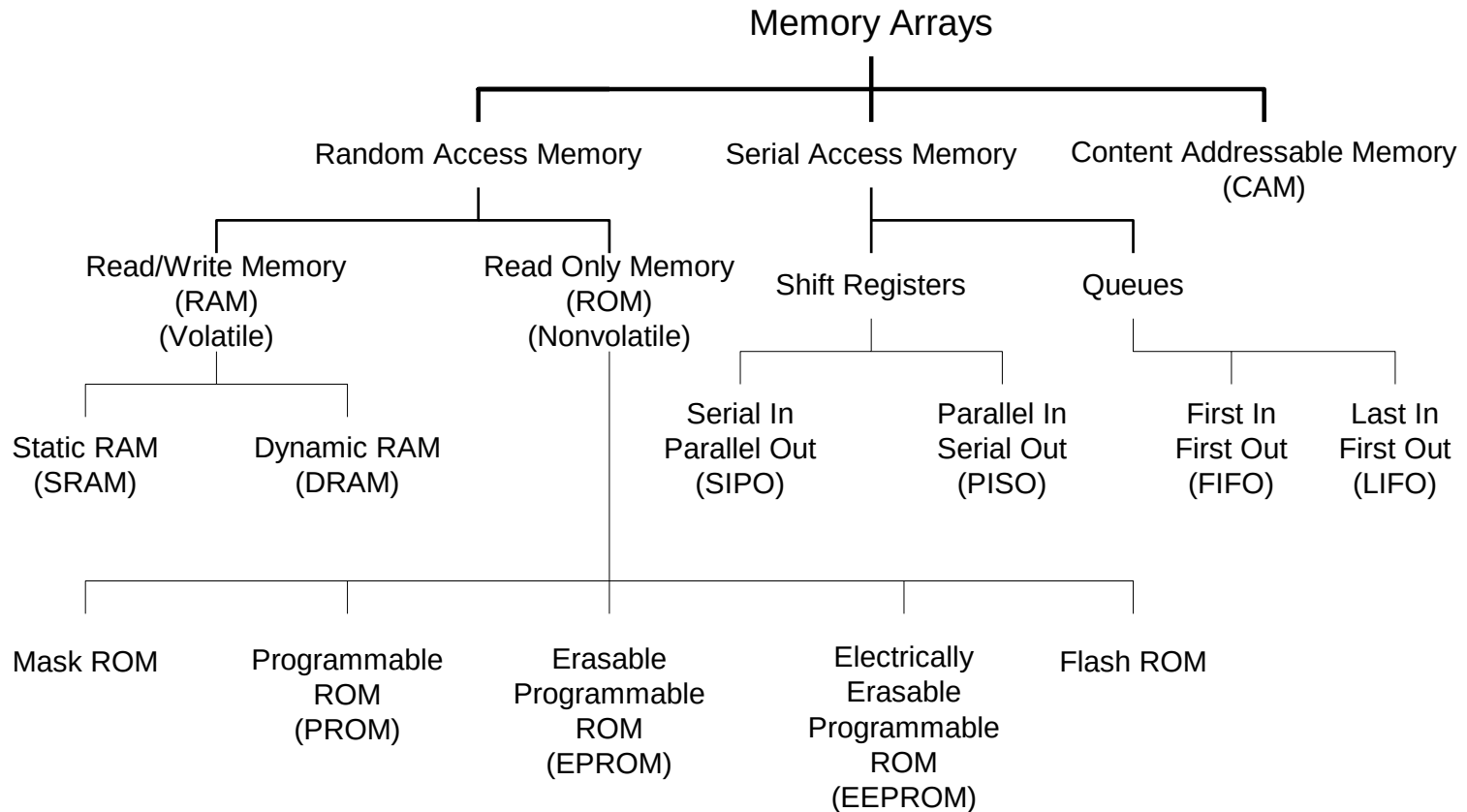




Array Subsystems

Memory Arrays



Lecture Content

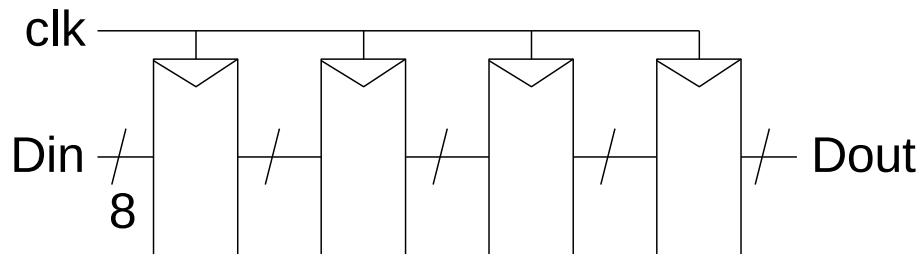
- ❑ Serial Access Memories
 - Shift Registers
 - Tapped Delay Lines
 - Serial In Parallel Out (SIPO)
 - Parallel In Serial Out (PISO)
 - Queues (FIFO, LIFO)

Serial Access Memories

- ❑ Serial access memories do not use an address
 - Shift Registers
 - Tapped Delay Lines
 - Serial In Parallel Out (SIPO)
 - Parallel In Serial Out (PISO)
 - Queues (FIFO, LIFO)

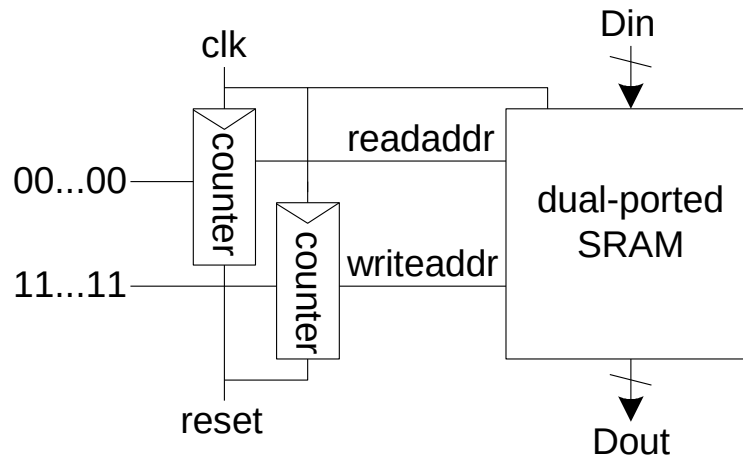
Shift Register

- ❑ *Shift registers* store and delay data
- ❑ Simple design: cascade of registers
 - Watch your hold times!



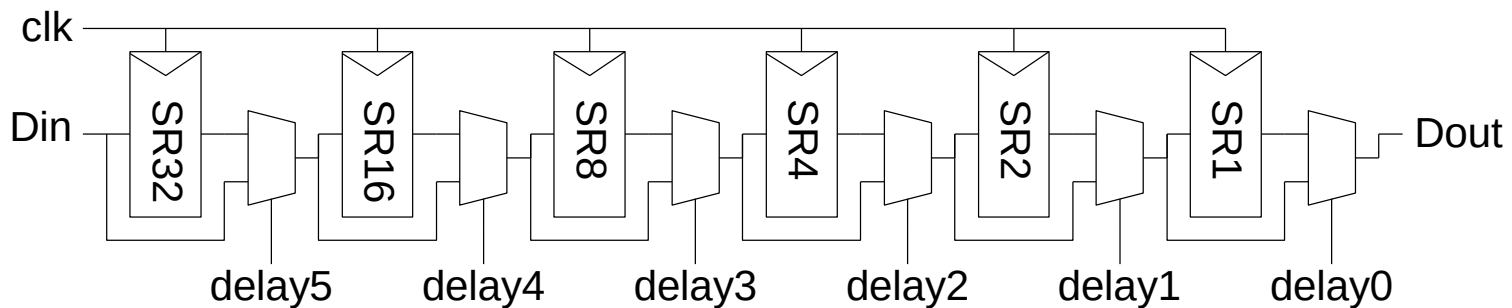
Denser Shift Registers

- ❑ Flip-flops aren't very area-efficient
- ❑ For large shift registers, keep data in SRAM instead
- ❑ Move read/write pointers to RAM rather than data
 - Initialize read address to first entry, write to last
 - Increment address on each cycle



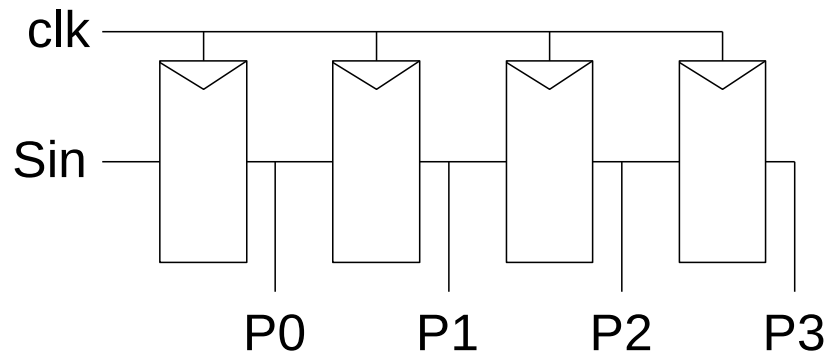
Tapped Delay Line

- ❑ A *tapped delay line* is a shift register with a programmable number of stages
- ❑ Set number of stages with delay controls to mux
 - Ex: 0 – 63 stages of delay



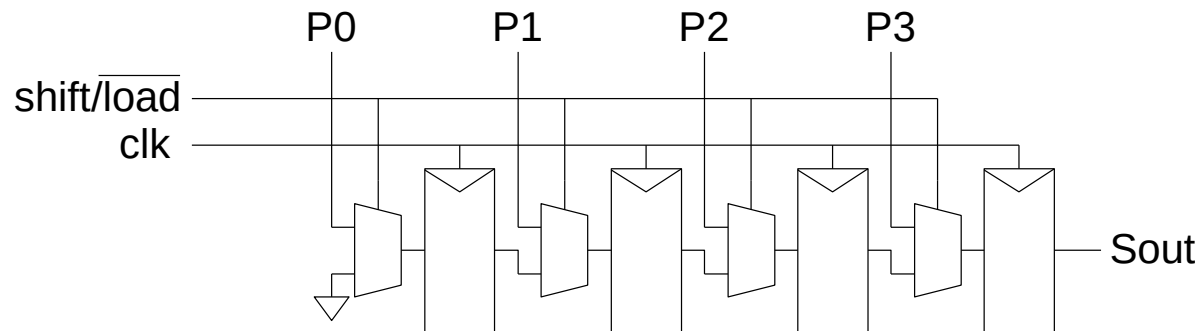
Serial In Parallel Out

- ❑ 1-bit shift register reads in serial data
 - After N steps, presents N-bit parallel output



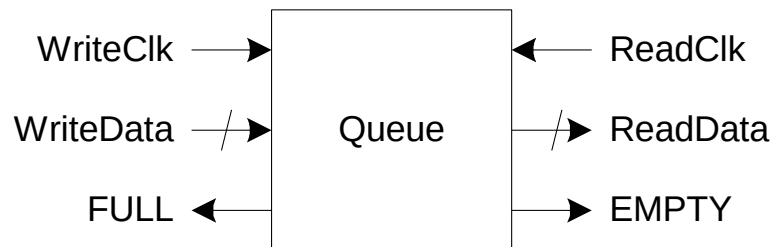
Parallel In Serial Out

- ❑ Load all N bits in parallel when shift = 0
 - Then shift one bit out per cycle



Queues

- ❑ *Queues* allow data to be read and written at different rates.
- ❑ Read and write each use their own clock, data
- ❑ Queue indicates whether it is full or empty
- ❑ Build with SRAM and read/write counters (pointers)



FIFO, LIFO Queues

❑ *First In First Out* (FIFO)

- Initialize read and write pointers to first element
- Queue is EMPTY
- On write, increment write pointer → FULL
- On read, increment read pointer → EMPTY

❑ *Last In First Out* (LIFO)

- Also called a *stack*
- Use a single *stack pointer* for read and write
- On write, the pointer is incremented → FULL
- On read, the pointer is decremented → EMPTY

Home Work

- ❑ Implement an 8-bit register file with 2 read ports and 1 write port at transistor level.